DATABASE SPECIFICATION | MCA SEMESTER 1 MINI PROJECT | 3NF STRICT RELATIONAL MODEL | v1.0 • Lossless Document

SkillConnect Data Dictionary

Relational Database Schema & Data Dictionary for Peer-to-Peer Student Skill Exchange Platform

RDBMS ENGINE	ORM FRAMEWORK	SCHEMA COVERAGE	AUTOMATED TESTS
SQLite 3 (WAL Mode)	**SQLAlchemy 2.0+ / Flask**	**7 Tables • 54 Attributes**	**89 / 89 Test Cases Passed**

1 Relational Schema Summary Matrix

#	TABLE NAME	ENTITY CLASSIFICATION	PK	FOREIGN KEYS	UNIQUE CONSTRAINTS	SYSTEM PURPOSE
1	`users`	Core / Identity Entity	`id`	—	email	Stores student and administrator credentials, profiles, academic institution, and role-based access permissions.
2	`skill_categories`	Taxonomy Entity	`id`	—	name	Standardized academic and technical taxonomy groupings for classifying platform skills into subject disciplines.
3	`skills`	Catalogue Entity	`id`	category_id	name	Canonical catalogue of predefined and student-proposed skills available for peer exchange.
4	`user_skills`	Associative / Junction Entity	`id`	user_id, skill_id	(user_id, skill_id, skill_type)	Junction table mapping student profiles to skills with teaching/learning intent and proficiency levels.
5	`exchange_requests`	Proposal / Workflow Entity	`id`	sender_id, receiver_id, offered_skill_id, requested_skill_id	—	Bilateral skill swap proposals initiated by one student to another with offered and requested skills.
6	`exchange_sessions`	Operational / Schedule Entity	`id`	request_id	request_id	Scheduled learning meetings originating from accepted exchange requests.
7	`ratings`	Evaluation / Review Entity	`id`	session_id, reviewer_id, reviewee_id	(session_id, reviewer_id)	Mutual post-session peer ratings and qualitative feedback submitted after completed learning sessions.

2 Normalized Table Specifications

1. users

10 Attributes | Core / Identity Entity

Stores student and administrator credentials, profiles, academic institution, and role-based access permissions.

COLUMN	DATA TYPE	CONSTRAINTS & KEYS	NULL	DEFAULT	DESCRIPTION & DOMAIN RULES
id	INTEGER	PK AUTOINC	NO	—	Unique primary identifier for user account
name	VARCHAR(120)	NOT NULL	NO	—	Full legal or display name of student / admin
email	VARCHAR(180)	UNIQUE INDEX	NO	—	Institutional email address (login credential)
password_hash	VARCHAR(255)	NOT NULL	NO	—	Cryptographically salted password hash (pbkdf2:sha256)
role	VARCHAR(20)	NOT NULL	NO	'student'	Authorization role: 'student' or 'admin'
college	VARCHAR(180)	NULLABLE	YES	—	Enrolled college / university institution name
bio	TEXT	NULLABLE	YES	—	Student biography, academic interests & background
availability	VARCHAR(120)	NULLABLE	YES	—	Preferred study hours (e.g. 'Weekends & Evenings')
created_at	DATETIME	NOT NULL	NO	UTC_NOW	Account creation timestamp (UTC)
updated_at	DATETIME	NOT NULL	NO	UTC_NOW	Profile last updated timestamp (UTC, onupdate)

Integrity & Relationships: One-to-Many: user_skills, sent_requests, received_requests, given_ratings, received_ratings.

2. skill_categories

5 Attributes | Taxonomy Entity

Standardized academic and technical taxonomy groupings for classifying platform skills into subject disciplines.

COLUMN	DATA TYPE	CONSTRAINTS & KEYS	NULL	DEFAULT	DESCRIPTION & DOMAIN RULES
id	INTEGER	PK AUTOINC	NO	—	Unique category primary identifier
name	VARCHAR(80)	UNIQUE NOT NULL	NO	—	Standardized category name (e.g. 'Programming & Tech', 'Database')
description	VARCHAR(255)	NULLABLE	YES	—	Overview of domain competencies covered in this category
icon	VARCHAR(50)	NOT NULL	NO	'bi-tag'	Bootstrap Icons CSS class for UI visual badge
created_at	DATETIME	NOT NULL	NO	UTC_NOW	Taxonomy creation timestamp (UTC)

Integrity & Relationships: One-to-Many: skills (cascade all, delete-orphan). Pre-seeded with 8 standard MCA disciplines.

3. skills

5 Attributes | Catalogue Entity

Canonical catalogue of predefined and student-proposed skills available for peer exchange.

COLUMN	DATA TYPE	CONSTRAINTS & KEYS	NULL	DEFAULT	DESCRIPTION & DOMAIN RULES
id	INTEGER	PK AUTOINC	NO	—	Unique skill primary identifier
category_id	INTEGER	FK NOT NULL	NO	—	Parent category ID referencing skill_categories(id)
name	VARCHAR(100)	UNIQUE INDEX	NO	—	Canonical skill name (e.g. 'Python', 'SQL', 'Figma', 'Docker')
description	VARCHAR(255)	NULLABLE	YES	—	Summary curriculum or domain details
created_at	DATETIME	NOT NULL	NO	UTC_NOW	Skill registration timestamp (UTC)

Integrity & Relationships: Many-to-One: skill_categories (category_id). One-to-Many: user_skills (cascade all, delete-orphan).

4. user_skills

8 Attributes | Associative / Junction Entity

Junction table mapping student profiles to skills with teaching/learning intent and proficiency levels.

COLUMN	DATA TYPE	CONSTRAINTS & KEYS	NULL	DEFAULT	DESCRIPTION & DOMAIN RULES
id	INTEGER	PK AUTOINC	NO	—	Unique junction record identifier
user_id	INTEGER	FK INDEX	NO	—	Student ID referencing users(id)
skill_id	INTEGER	FK INDEX	NO	—	Skill ID referencing skills(id)
skill_type	VARCHAR(10)	NOT NULL	NO	—	Intent type: 'teach' (offering) or 'learn' (seeking)
proficiency	VARCHAR(20)	NOT NULL	NO	'intermediate'	Proficiency: 'beginner', 'intermediate', 'advanced'
notes	VARCHAR(255)	NULLABLE	YES	—	Custom student notes, target frameworks, or topic focus
created_at	DATETIME	NOT NULL	NO	UTC_NOW	Record addition timestamp (UTC)
updated_at	DATETIME	NOT NULL	NO	UTC_NOW	Record last modified timestamp (UTC, onupdate)

Integrity & Relationships: Foreign Keys: user_id -> users(id), skill_id -> skills(id). Table Unique Constraint: uq_user_skill_type.

5. exchange_requests

9 Attributes | Proposal / Workflow Entity

Bilateral skill swap proposals initiated by one student to another with offered and requested skills.

COLUMN	DATA TYPE	CONSTRAINTS & KEYS	NULL	DEFAULT	DESCRIPTION & DOMAIN RULES
id	INTEGER	PK AUTOINC	NO	—	Unique exchange proposal identifier
sender_id	INTEGER	FK INDEX	NO	—	Initiating student ID referencing users(id)
receiver_id	INTEGER	FK INDEX	NO	—	Recipient peer student ID referencing users(id)
offered_skill_id	INTEGER	FK NOT NULL	NO	—	Skill sender teaches, referencing skills(id)
requested_skill_id	INTEGER	FK NOT NULL	NO	—	Skill sender wants to learn, referencing skills(id)
status	VARCHAR(20)	NOT NULL	NO	'pending'	State: 'pending', 'accepted', 'rejected', 'cancelled'
message	TEXT	NULLABLE	YES	—	Introductory proposal message and schedule preferences
created_at	DATETIME	NOT NULL	NO	UTC_NOW	Proposal sent timestamp (UTC)
updated_at	DATETIME	NOT NULL	NO	UTC_NOW	Status transition timestamp (UTC, onupdate)

Integrity & Relationships: Foreign Keys: sender_id, receiver_id -> users(id); offered_skill_id, requested_skill_id -> skills(id).

6. exchange_sessions

10 Attributes | Operational / Schedule Entity

Scheduled learning meetings originating from accepted exchange requests.

COLUMN	DATA TYPE	CONSTRAINTS & KEYS	NULL	DEFAULT	DESCRIPTION & DOMAIN RULES
id	INTEGER	PK AUTOINC	NO	—	Unique session identifier
request_id	INTEGER	FK UNIQUE	NO	—	Parent accepted request ID referencing exchange_requests(id)
scheduled_at	DATETIME	NOT NULL	NO	—	Date and time of meeting
duration_minutes	INTEGER	NOT NULL	NO	60	Planned meeting length in minutes (default 60)
mode	VARCHAR(20)	NOT NULL	NO	'online'	Delivery format: 'online', 'in_person', 'hybrid'
meeting_details	VARCHAR(255)	NULLABLE	YES	—	Video call URL (Meet/Zoom) or physical campus room
notes	TEXT	NULLABLE	YES	—	Session agenda, study materials, or preparation topics
status	VARCHAR(20)	NOT NULL	NO	'scheduled'	Session state: 'scheduled', 'completed', 'cancelled'
created_at	DATETIME	NOT NULL	NO	UTC_NOW	Session booking timestamp (UTC)
updated_at	DATETIME	NOT NULL	NO	UTC_NOW	Session modification timestamp (UTC, onupdate)

Integrity & Relationships: Foreign Key: request_id -> exchange_requests(id) [UNIQUE 1-to-1]. One-to-Many: ratings.

7. ratings

7 Attributes | Evaluation / Review Entity

Mutual post-session peer ratings and qualitative feedback submitted after completed learning sessions.

COLUMN	DATA TYPE	CONSTRAINTS & KEYS	NULL	DEFAULT	DESCRIPTION & DOMAIN RULES
id	INTEGER	PK AUTOINC	NO	—	Unique evaluation record identifier
session_id	INTEGER	FK INDEX	NO	—	Completed session ID referencing exchange_sessions(id)
reviewer_id	INTEGER	FK INDEX	NO	—	Student submitting review, referencing users(id)
reviewee_id	INTEGER	FK INDEX	NO	—	Student receiving review, referencing users(id)
score	INTEGER	CHECK 1..5	NO	—	Integer rating between 1 and 5 stars (inclusive)
feedback	TEXT	NULLABLE	YES	—	Qualitative peer feedback, testimonials & remarks
created_at	DATETIME	NOT NULL	NO	UTC_NOW	Rating submission timestamp (UTC)

Integrity & Relationships: Foreign Keys: session_id -> exchange_sessions(id), reviewer_id, reviewee_id -> users(id).

3 Relational Integrity, Cascade & Business Rules

CONSTRAINT **Unique Intent Constraint**

`UNIQUE(user_id, skill_id, skill_type)` on `user_skills` guarantees a student cannot create duplicate portfolio records for the same skill and intent combination.

1-TO-1 **Strict Session 1-to-1 Mapping**

`UNIQUE(request_id)` on `exchange_sessions` enforces that each accepted proposal generates exactly one scheduled learning meeting.

CHECK **Star Rating Boundaries & Uniqueness**

`CHECK(score BETWEEN 1 AND 5)` enforces valid 1-to-5 star ratings, while `UNIQUE(session_id, reviewer_id)` prevents duplicate evaluations from the same participant.

INTEGRITY **Referential Cascades & Self-Check**

Deleting a `User` cascades across related skills, proposals, and reviews. Business logic enforces `sender_id != receiver_id` to forbid self-swaps.